



tive, prompt engineering focuses on designing and developing prompts. In 2022 prompt engineering gained prominence with the launch of new machine-learning models. Models like Stable Diffusion and DALL-E have been the foundations of text-to-image prompting. These models can convert text prompts into visuals, as the name implies.

Stable Diffusion and DALL-E require detailed descriptions of desired visual output as prompts. In contrast, prompts for large language models (LLMs) like ChatGPT and GPT-3 can be simple queries, or they can involve complex problems and instructions with multiple facts. You can also use random statements like “Tell me the description” for a specific prompt. In prompt engineering examples, it’s important to highlight the essential components of a prompt. At a high level, a prompt consists of four distinct components: instructions, questions, input data, and examples. Effective prompts require an appropriate combination of these elements to achieve successful results.

Principles of Prompt Engineering

Prompts are an essential element of prompting and demonstrate the ease of training AI models to perform desired tasks. If you’re curious about how to design prompts, there are important principles that can guide their development for AI models.

Generation of useful outputs

A fundamental focus in a prompt engineering tutorial is guiding AI models to generate useful outputs. Consider a scenario where you need a summary of an article. To train a large language model with enough data to obtain a summary, you would use a prompt. The prompt would include the input text and a description of the task, outlining how you want the summary to be generated.

Variations of prompts

To achieve the best results in prompt engineering, it’s important to work on different variations of prompts. Even slight variations in the design of a prompt can generate significantly different outputs. AI models learn that different formulations of a prompt are applicable for different contexts and purposes.

Likelihood features

Likelihood features can also be useful in prompt engineering. They help identify whether the model has difficulty understanding specific words, structures, or phrases. It’s important to note that likelihood is typically higher in the initial stages of the sequence, and the model may assign a low likelihood to a new prompt before becoming familiar with it over multiple iterations.

Defining tasks

Defining tasks in prompt engineering involves including additional components in the task description. Providing adequate context to the model can generate desired results in prompt engineering applications. The critical components of prompts, such as input and output indicators, help to clearly define the desired task to the model. A detailed description of prompt components is particularly useful when including multiple examples in a prompt.

Overview of prompt patterns

Classic software patterns serve the same purpose as prompt patterns and are written in the same way, with a few small changes to fit the context of output creation with LLMs. Below is a summary of different prompt patterns you can use:

1. **A name and a category:** The name of the question pattern is the only way to identify it and shows what problem is being solved.
2. **Intention and setting:** The problem the prompt pattern solves and the goals it helps you reach are described in the purpose and context. The problem should ideally not be tied to a specific domain, but patterns that are tied to a specific domain can be recorded with a review of the situation where the pattern applies.
3. **Reason:** The reason shows why the problem exists and why it’s important to solve it. The motivation is described in terms of how people connect with a conversational LLM and how it can be better than when people just ask the LLM to do something. Documents list the specific situations where the changes are supposed to happen.
4. **The context.** The cue pattern gives the LLM basic background information in the form of a list of key ideas, which are described by the structure. Like “participants” in a programme scheme, these ideas are important to the whole. The surrounding information can be communicated in different ways, just like a software pattern can be implemented in different ways in code. However, the most important information that makes up the pattern should be communicated.
5. **Example implementation:** An example shows how the prompt pattern is written in real life and what kinds of results an LLM makes.
6. **Consequences:** These list the pros and cons of using the pattern and may help you figure out how to change the prompt for different situations.

Careers in Prompt Engineering

As the applications of prompting continue to expand, so do the